



Lab 13

CMOS Amplifier Design

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13.1 Objective

To get familiarized with CMOS circuits and implement CMOS common source amplifier with active load.

Task 1: Pre-Lab

Explain the construction, working, and modes of operation of a PMOS. Why NMOS is faster than PMOS?

A PMOS transistor (P-channel Metal-Oxide-Semiconductor) is a type of MOSFET where the majority carriers are holes. It has three terminals: Source (S), Drain (D), and Gate (G). In a PMOS, the source and drain are made of p-type material, while the substrate (body) is n-type. An insulating layer of silicon dioxide separates the gate from the channel.

In normal operation, the source is at a higher voltage than the drain. When a negative voltage is applied to the gate relative to the source, it attracts holes to the channel region, forming a p-type channel between source and drain. This allows current to flow from source to drain. If the gate is at the same or higher voltage than the source, the channel disappears and the PMOS remains off.

There are two main modes of operation:

1. **Cut-off Region** – When gate-to-source voltage (V_{GS}) is greater than or equal to the threshold voltage (V_{th}), the transistor is off and no current flows.
2. **Saturation and Linear (Ohmic) Region** – When V_{GS} is less than V_{th} (i.e., sufficiently negative), the PMOS turns on. The linear region occurs when drain-to-source voltage is small; the saturation region occurs when this voltage is large enough for the channel to pinch off at the drain end.

NMOS transistors are faster than PMOS because electrons, which are the charge carriers in NMOS, have higher mobility than holes, which are the carriers in PMOS. Higher mobility means that electrons can move faster through the channel, resulting in quicker switching and better speed performance. This is why modern CMOS designs prefer NMOS for performance-critical paths and often use NMOS transistors more heavily in logic circuits.

13.2 Introduction to CMOS Circuits

CMOS technology is the dominant fabrication technology used for electronic systems including computers, smart-phones, TVs and other household appliances. CMOS is short for Complementary Metal Oxide Semi-conductor and it is a technology used for fabricating integrated circuits using both

PMOS and NMOS transistors. The combination of PMOS and NMOS makes it easy to design circuits at scaled-down size. CMOS circuits are the most widely used IC technology for both analog and digital as well as combined analog and digital (or mixed-signal) applications. Figure 1 shows CMOS device structure where 2 transistor types are combined in one structure.

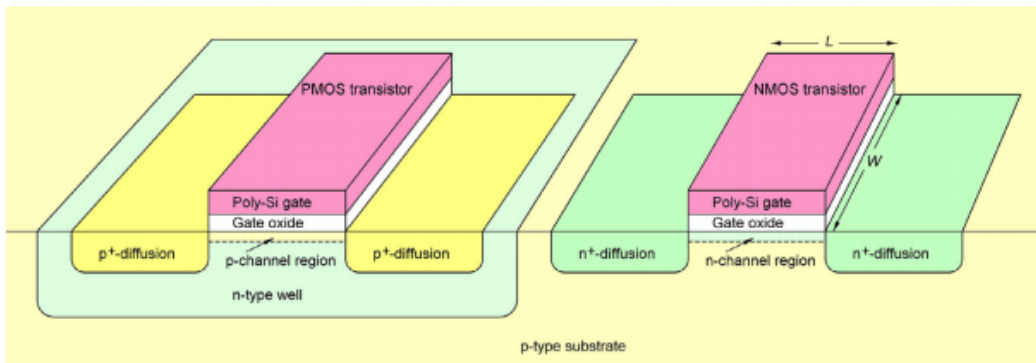


Figure 1: CMOS device structure

13.3 CMOS Amplifier

As the name implies, a MOS based amplifier that utilizes both; n-channel and p-channel MOSFETs is known as CMOS amplifier. Common Source (CS) configuration of single stage MOSFET amplifier can offer substantial voltage gain in combination with large input impedance. CMOS implementation of common-source MOS amplifiers is used in integrated circuits for amplification.

Figure 2 represents a basic CS amplifier circuit with resistive load R_D . We have implemented the following circuits of common-source amplifier in previous labs:

- 1) Common-source amplifier with resistive load R_D and voltage-divider bias
- 2) Common-source amplifier with diode load (R_D replaced by diode-connected NMOS)
- 3) Common-source amplifier with triode load (R_D replaced by PMOS operating in triode region as resistor)
- 4) Common-source amplifier (with resistive load R_D) biased by MOS constant-current source

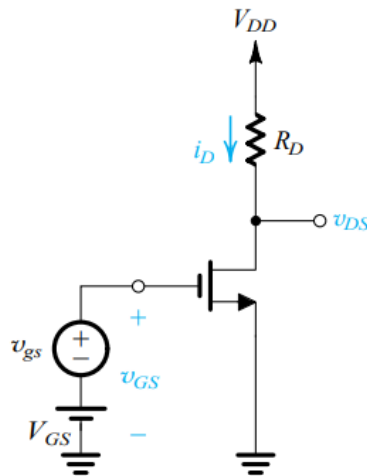


Figure 2: CS amplifier with resistive load and fixed bias with amplifier gain $A_v = -gm(R_D || r_o)$

CS amplifier with resistive loads provide limited gain as the gain of amplifier is dependent on R_D for given DC bias operating conditions. Increasing R_D can push the MOS amplifier from saturation to triode region. Moreover, in integrated circuit design, a resistor normally takes up more Silicon area than transistors. The triode load requires precise gate voltage at load MOS and CS amplifier with diode connected MOS load requires large difference in aspect ratio to achieve amplifier gain.

In this lab, we are going to replace the MOS load in CS amplifier by a constant current-source (implemented by MOSFETs). Such a configuration is called MOS common-source amplifier with **Active Load** or **Current-Source Load**.

13.4 MOS Common-Source Amplifier with Current-Source Load

The circuit shown in Figure 3(a) represents a CS amplifier circuit with current-source load. A practical current source can be replaced by constant-current source with a resistance in parallel as shown in Figure 3 (b).

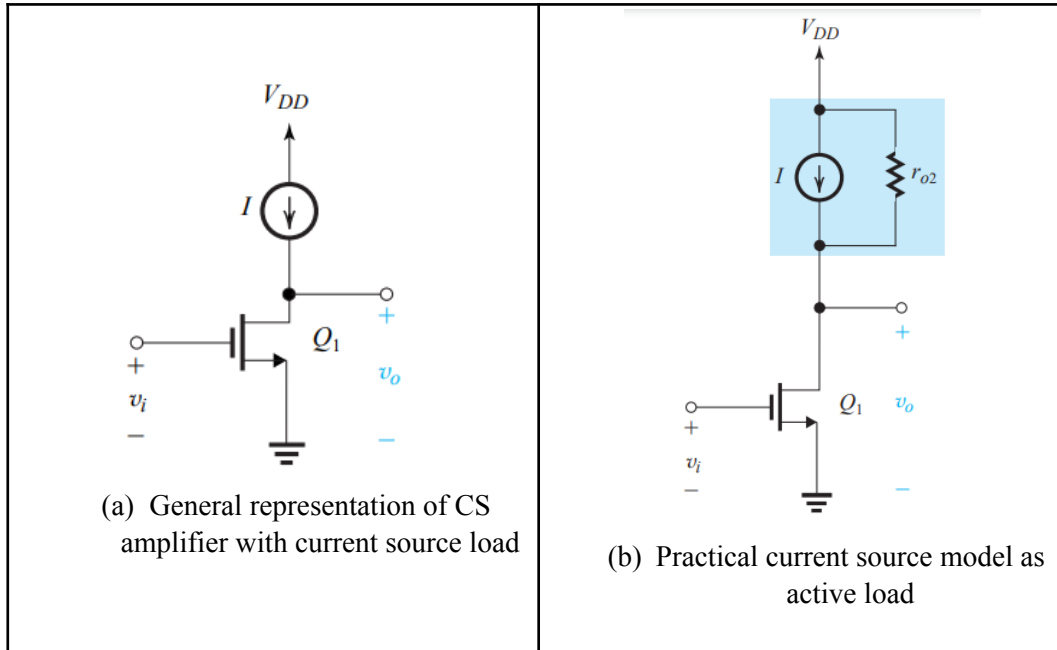


Figure 3: CS amplifier with active load (current source)

13.5 Small Signal Analysis of Common-Source Amplifier with Current-Source Load

The resistance of current-source is offered to the CS amplifier MOSFET in place of resistive load R_D . In small signal analysis, the DC current source will be replaced by open-circuit and source resistance will be in parallel to the MOS output resistance as shown in Figure 4 which is the small-signal model of CS amplifier circuit with active load of Figure 3(b).

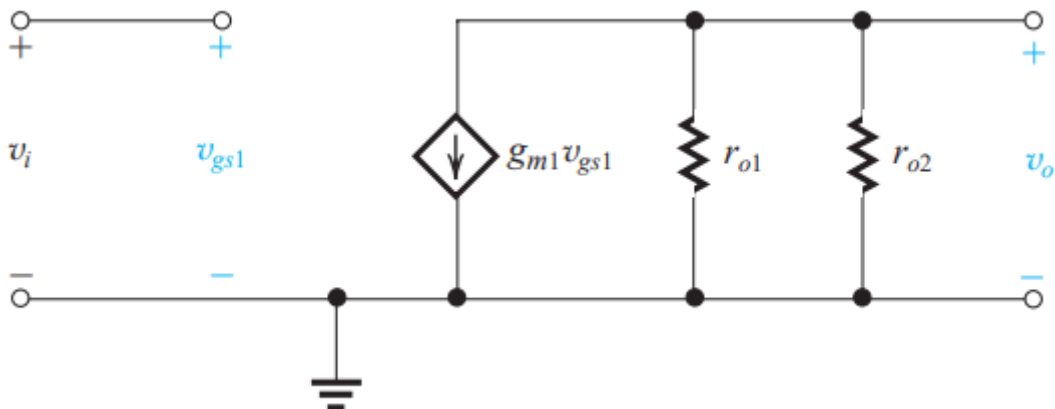


Figure 4: Small-signal model of CS amplifier with active load

In comparison to the load resistor R_D , the resistance of current-source is quite large. The gain of common-source amplifier with current-source load can be written as;

$$A_v = -g_m(r_{o1} \parallel r_{o2})$$

This is called intrinsic gain of the amplifier.

13.6 Current Source Circuit by p-channel MOSFETs

The constant-current source and current mirror circuits implemented in last lab using NMOS transistors can be implemented using PMOS transistors. Figure 5 shows implementation of a current-source circuit using PMOS. Here again, the drain and gate terminals of Q1 are tied together that keep the MOS in saturation region and resistance R is used to establish reference current. This current is mirrored by other PMOS Q2 whose gate terminal is tied to the gate of MOS Q1. Note that for PMOS circuits, source terminal is connected to the positive supply V_{DD} .

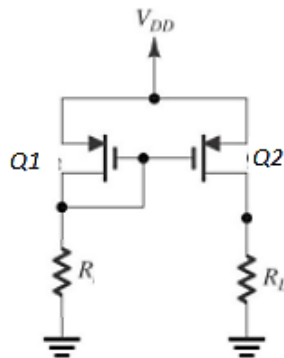


Figure 5: PMOS current mirror circuit

Task 2: To investigate the output resistance of p-channel MOS due to channel length modulation through characteristic curve

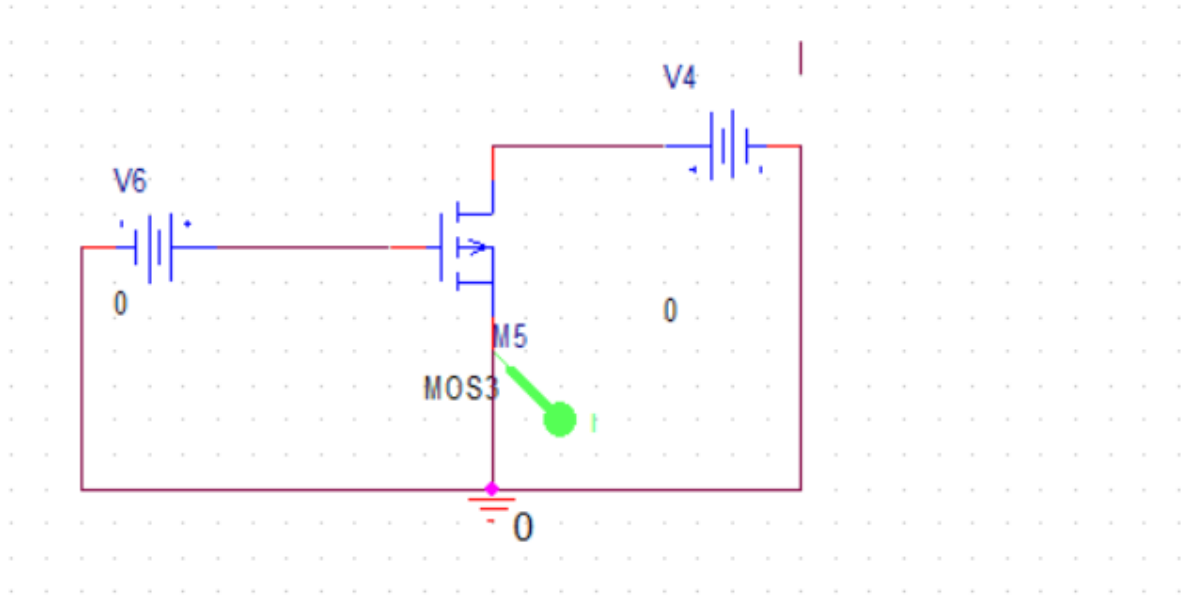
1. Add a PMOS model in PSpice using MbreakP3. Edit the model to state the following given specifications of model:
 - Threshold voltage: -1V
 - Process trans-conductance: 0.02m A/V²
 - Length: 0.25 um
 - Width: 6.25 um
 - Channel-length modulation coefficient: 0.01 V⁻¹
2. To obtain its characteristic curve, sweep the drain to source voltage from -10V to 0V for different values of gate to source voltage (-1, -2, -3 V). Measure current from source to drain.
3. For each of the obtained curves, find the value of output resistance r_o in saturation region using the formula given below. Complete Table 1.

$$r_o = \left. \frac{\partial V_{DS}}{\partial I_D} \right|_{V_{GS}=\text{constant}} = \left. \frac{\Delta V_{DS}}{\Delta I_D} \right|_{V_{GS}=\text{constant}}$$

4. This output resistance can be estimated by channel-length modulation coefficient. Estimate the output resistance using value of drain current (in middle of saturation region) and specified channel-length modulation coefficient.

$$r_o = \frac{1}{\lambda I_D}$$

Circuit:



WaveForm:

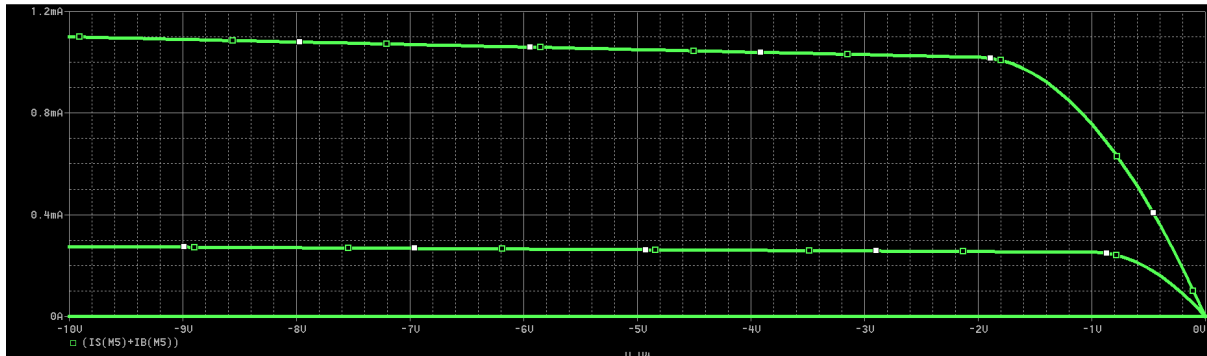


Table 1: Output resistance of MOSFET with channel-length modulation incorporated (swap the values)

Gate-to-Source Voltage V_{GS}	Output Resistance: r_o	Estimated Output Resistance : r_o
-2V	initial vds = -6 final vds = -7 initial id = 265u final id = 267.5	lambda = 0.01 id = 265u
-3 V	initial vds = -6 final vds = -7 initial id = 1.06mA final id = 1.07mA	id = 1.06mA lambda = 0.01

Task 3: To design PMOS current-source for given reference current

- For implementing the circuit shown in Figure 5, use the PMOS model created in Task 2 for Q1 and Q2.
- To design the current-source for obtaining 540uA current at output, find the value of resistance R. $V_{DD}=10V$. You can use the following equations for PMOS;

$$I_D = \frac{1}{2} k_n' \frac{W}{L} (V_{SG} - |V_t|)^2$$

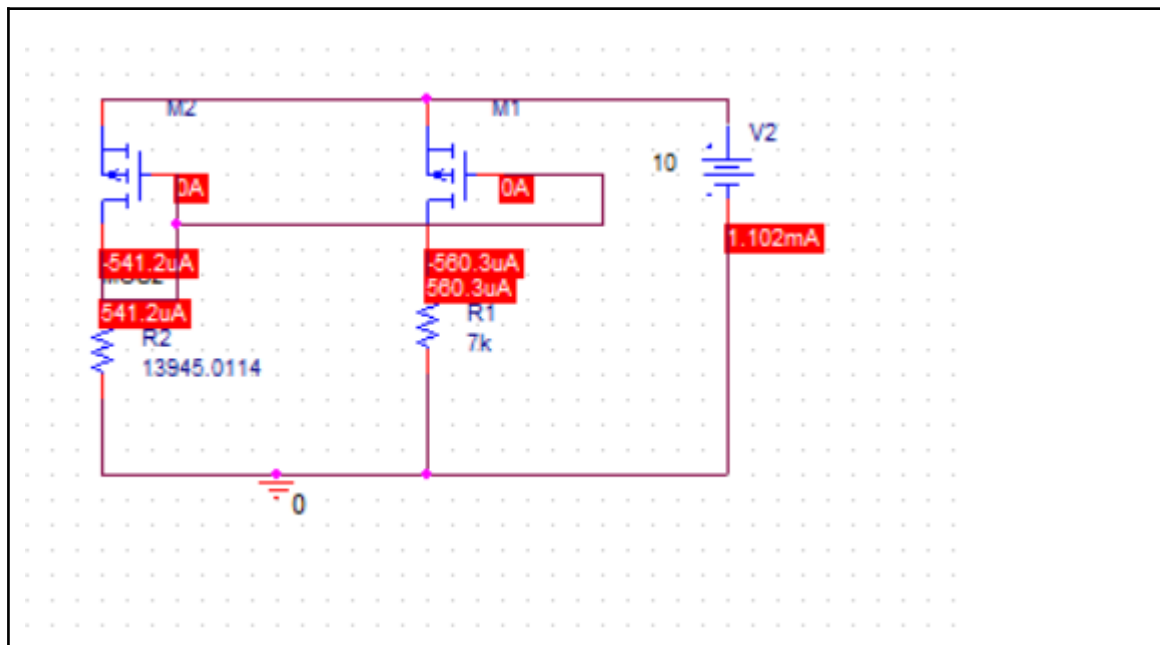
$$V_{SD} = V_{DD} - I_D R$$

3. Simulate the above circuit to verify your design. Connect a 7k Ohms resistor at output.

Table 1: PMOS Current Source

Calculated Value of R	13945.0114 ohm
Simulated Reference Current	541.uA
Simulated Output Current	560.3uA

4. Add captures of the implemented circuit and result.



5. Compare the output and reference current. If output current is different from the reference currents, what is the possible reason of any discrepancies in the results?

The primary reason is the channel length modulation effect, which the theoretical calculation ignores but the simulation incorporates.

13.7 Common-Source Amplifier Circuit with Current-Source Load and Current-Source Biasing

The circuit shown in Figure 6 is CS amplifier loaded with PMOS current source. This current source is made of basic PMOS current mirror (with M4 and M5). The output impedance for the PMOS current mirror is r_{0M5} which serves as a load in small signal equivalent circuit for gain calculations.

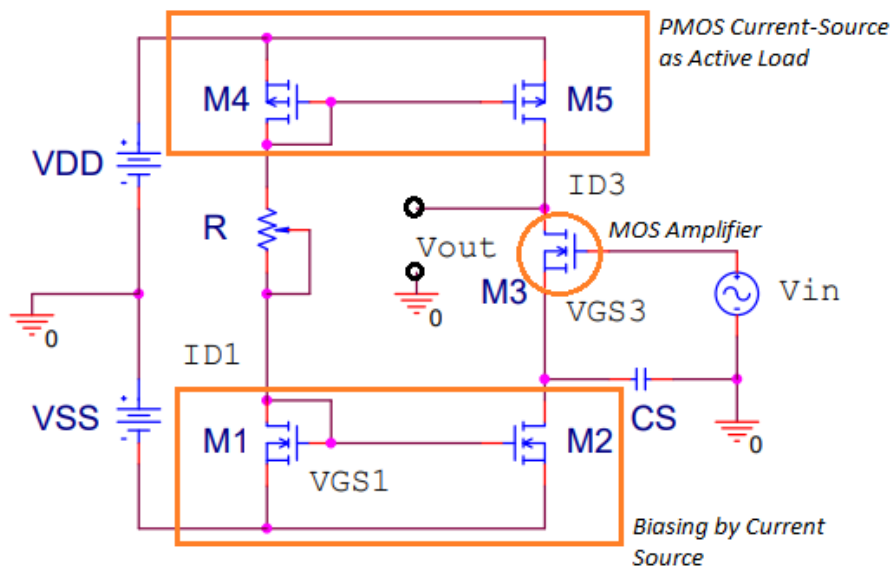


Figure 6: CMOS CS amplifier (CS amplifier with PMOS current-source with current-source biasing)

The gain of the CS amplifier with active load becomes:

$$A_v = -g_m(r_{OM5} \parallel r_o)$$

The pair of M1 and M2 is used for biasing of amplifier transistor M3. The value of R has to be adjusted to establish reference current for current-source load and biasing.

Task 4: To implement common-source amplifier with PMOS current-source load

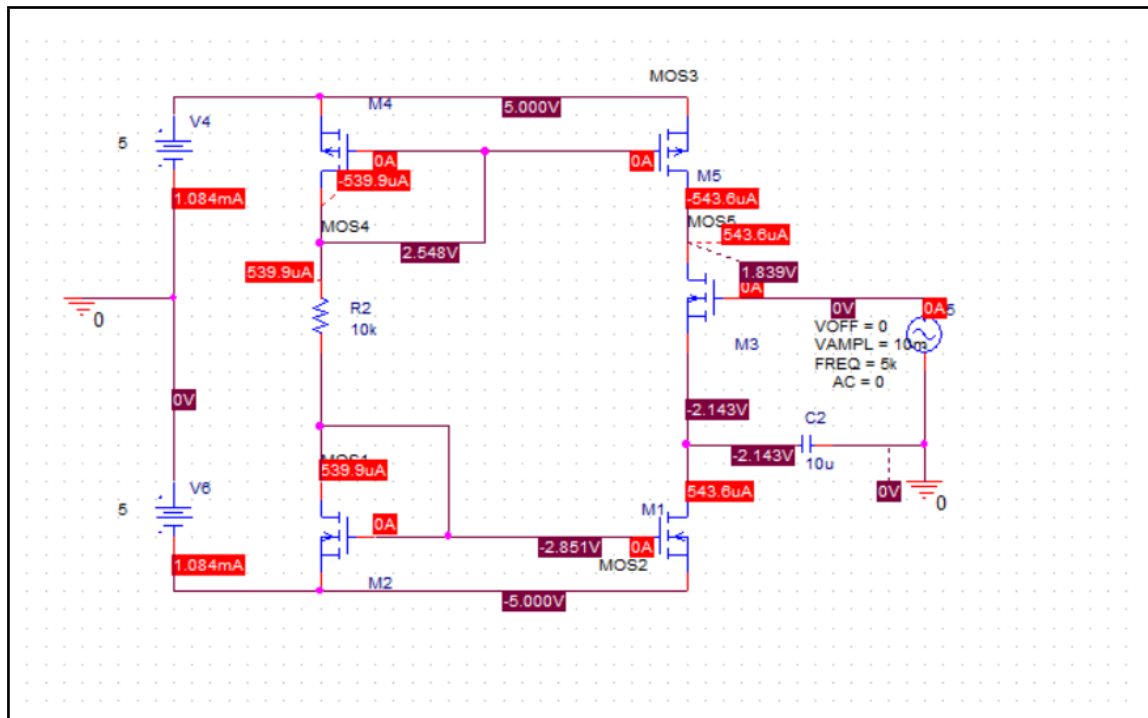
1. Implement the CS amplifier circuit shown in Figure 6. Implement the current-source load using PMOS models used in previous tasks for M4 and M5. The NMOS transistors (M1 and M2) used for biasing and the amplifier MOS (M3) have following specifications:

- Threshold voltage: 1V
- Process trans-conductance: 0.8m A/V²
- Length: 1 um
- Width: 1 um

- Channel-length modulation coefficient: 0.01 V^{-1}
2. The value of R selected to be $10\text{k}\Omega$. $V_{SS} = V_{DD} = 5\text{V}$.

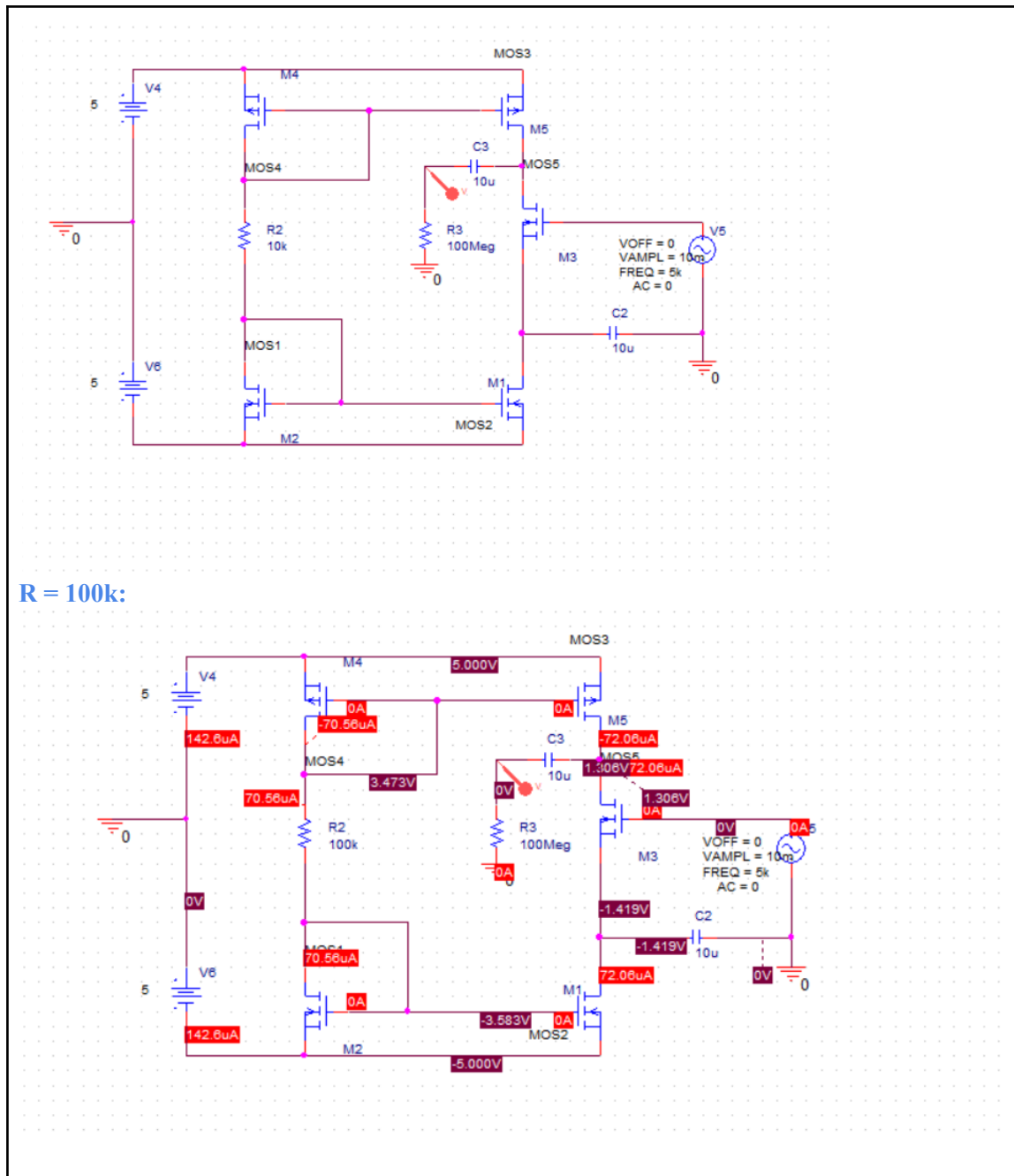
Task 5: To investigate the characteristics of common-source amplifier with PMOS current-source load

1. Perform DC analysis (through simulation) to obtain the values listed in Table 2.



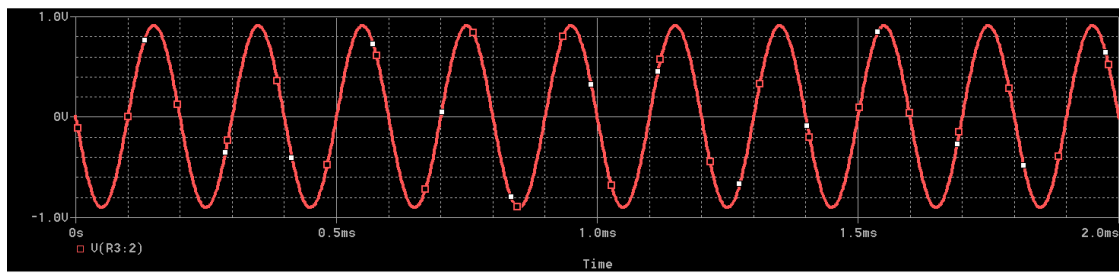
2. Apply AC sinusoid input of 10mV amplitude 5 kHz at input of the amplifier. Measure unloaded output voltage of amplifier (to simulate open circuit at amplifier output terminals, we connect large resistance 100meg Ω in PSpice). Also, connect another coupling capacitor of 10uF with to measure output.

r = 10k

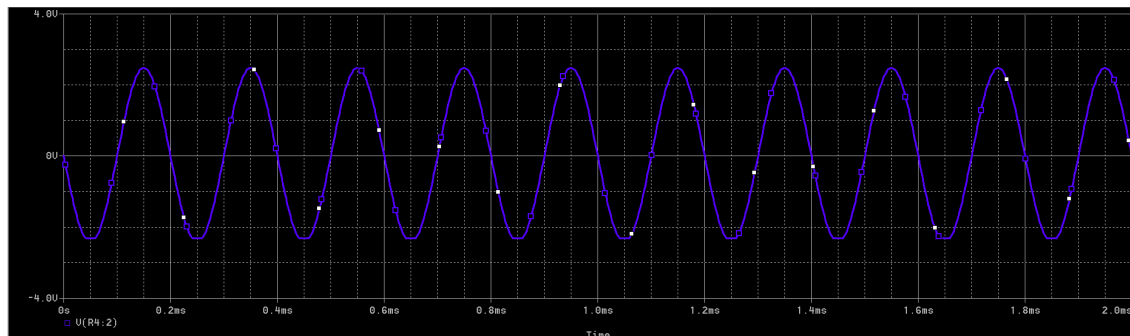


- Now, change the resistance used to set reference current from $10k\Omega$ to $100k\Omega$. Measure the reference current and drain current of MOS amplifier now. What changes do you observe in gain of amplifier?

Waveform 10k:



Waveform 100k:



Observation:

There's no phase change or any kind of inversion in the signal but just that the amplitude has been doubled by a gain of 2.

4. Using the formula for intrinsic gain of amplifier, try to relate the gain with drain current by using the expressions for r_o in terms of drain current and $g_m = \sqrt{2I_D K n' \left(\frac{W}{L}\right)}$ (note that this formula depends on MOS geometry and drain current).

The gain decreases as I_D increases. For example, when the reference current decreases (e.g., increasing R from 10kΩ to 100kΩ), I_D reduces, leading to a higher gain. This matches the lab observation where increasing R resulted in higher measured gain.

5. Explain why the amplifier is called CMOS amplifier? Why do you think this configuration is preferred in IC design?

High Gain: Active loads (PMOS current mirrors) provide high output impedance (r_o), boosting voltage gain.

Area Efficiency: Transistors replace bulky resistors, saving silicon area.

Low Power Consumption: CMOS technology minimizes static power dissipation.

Scalability: Compatible with modern nanometer fabrication processes for dense integration. This configuration is widely preferred in analog ICs for its performance and compatibility with digital CMOS processes.

Table 2: CMOS CS amplifier parameters

Parameters	Values
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Drain Current M3: I_D	543.6uA
Drain Voltage M3: V_D	1.839V
Source Voltage M3: V_S	-2.143V
Drain-to-Source Voltage M3: V_{DS}	3.982V
Gate-to-Source Voltage M3: V_{GS}	2.143
Trans-conductance M3: $g_m = \frac{2I_D}{V_{GS} - V_T} = \sqrt{2I_D Kn'(\frac{W}{L})}$	9.326×10^{-4}
Output Resistance M3: $r_o = \frac{1}{\lambda I_D}$	183958.79 ohm
Output Resistance of Load M5: r_{o5}	183958.79 ohm
Calculated Gain: $A_v = -g_m(r_{o5} r_o)$	-85.779
Mode of Operation of M3	saturation
Measured Gain: A_v	90.9852

13.8 Post Lab Task

Task 6: To explore IC Fabrication: From Sand to Chip

1. Explain the fabrication of integrated-circuit chips from Sand (Silicon) and provide a concise summary (no more than 200 words) about it.

Integrated circuits (ICs) begin with silicon, derived from purified sand (silica, SiO_2). The silicon undergoes refinement to achieve electronic-grade purity (99.9999%) via the Siemens process. This high-purity silicon is melted and grown into single-crystal ingots (boules) using the Czochralski method. The boules are sliced into thin wafers, polished to a flawless finish.

Fabrication involves photolithography: a wafer is coated with photoresist, exposed to UV light through a patterned mask, and developed to etch circuit layers. Repeated cycles of doping (ion implantation/diffusion) create n-type/p-type regions, while etching (wet/dry) removes excess material. Insulating (SiO_2) and conductive (Al, Cu) layers are deposited via CVD or PVD.

After layering, the wafer is tested for defects. Functional dies are diced from the wafer, packaged into protective casings with electrical connections, and rigorously tested for performance. Key advantages include scalability, precision, and compatibility with nanoscale processes.



Silicon, extracted from sand, is purified into electronic-grade crystals. These are sliced into wafers and patterned via photolithography. Repeated doping, etching, and deposition build transistors and interconnects. Finished wafers are diced, packaged, and tested. CMOS technology's efficiency, low power, and miniaturization make it ideal for modern ICs, enabling devices like smartphones and computers.

Assessment Rubric
Lab 13
CMOS Amplifier Design

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Points Distribution

Task No.	LR2 Simulation 32	LR 4 Data Collection 20	LR 5 Plots 12	LR 6 Calculatio n 16	LR 10 Analysis 16	CLO Mapped
Task 1	-	-	-	-	/4	CLO 1
Task 2	/8	/4	/4	/4	-	CLO 1
Task 3	/8	/4	-	/4	/4	CLO 2
Task 4	/12	-	-	-	-	CLO 3
Task 5	/4	/12	/8	/8	/4	CLO 1
Task 6	-	-	-	-	/4	CLO 1
	/32	/20	/12	/16	/16	

Affective Domain Rubric		Points	CLO Mapped
AR 4	Report Submission	4	CLO 4

CLO	Total Points	Points Obtained
1	64	
2	20	
3	12	
4	4	
Total	100	

For description of different levels of the mapped rubrics, please refer the provided Lab Evaluation Assessment Rubrics and Affective Domain Assessment Rubrics.